

Functional Specification Document (FSD)

Automated IT Procurement: Standard Laptop Orders via Flow Designer

Document Information

- **Document Identifier:** FSD-ITSM-PROC-2026-V1
- **Project Name:** Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
- **Target Platform:** ServiceNow Washington DC / Vancouver Enterprise Platform
- **Document Owner:** Lead Technical Architect & Systems Engineering
- **Current Version:** 1.0 (Production Sign-Off)
- **Publication Date:** 2026-09-30
- **Classification:** Internal Enterprise Technical Baseline

1. Document Control & Revision History

1.1 Revision History

Version	Release Date	Author / Role	Summary of Changes	Approval Status
0.1	2026-08-15	Lead Business Analyst	Initial draft capturing functional requirements and process pain points	Draft
0.5	2026-09-02	ServiceNow Solutions Architect	Technical architecture definition, Flow Designer action map, and schema definitions	Reviewed
0.9	2026-09-20	Lead Implementation Engineer	Incorporation of UAT test cases, UI policies, client scripts, and error-handling framework	Baseline
1.0	2026-09-30	Phase 6 Documentation Lead	Final production baseline incorporating UAT results, sign-offs, and operational runbook	Approved

1.2 Document Approvals & Sign-Off Matrix

Role	Approver Name	Title / Organization	Signature Status	Sign-Off Date
Project Sponsor	Jonathan Reynolds	Vice President, Global IT Infrastructure	APPROVED	2026-09-30
Lead Enterprise Architect	Dr. Aris Thorne	Principal Enterprise Architect	APPROVED	2026-09-30

ITSM Platform Owner	Kevin Zhang	Lead ServiceNow Architect	APPROVED	2026-09-30
Quality Assurance Lead	Priya Patel	Head of Enterprise QA & Test Engineering	APPROVED	2026-09-30
IT Procurement Operations	Derek Vance	Director of IT Procurement & Asset Management	APPROVED	2026-09-30
Information Security (SecOps)	Sarah Jenkins	Chief Information Security Officer (CISO Delegate)	APPROVED	2026-09-30

1.3 Referenced Architectural & Governance Baselines

- 1. Enterprise ITIL v4 Service Management Architecture Standard (Doc Ref: ITSM-STD-2025-V2)
- 2. ServiceNow Flow Designer Best Practice & Style Guide (SN-BP-FD-04)
- 3. Corporate Information Security Access Control Policy (SEC-POL-RBAC-09)
- 4. Hardware Lifecycle & Hardware Asset Management (HAM) Operational Policy (HAM-POL-2024)

2. Executive Summary & Project Purpose

2.1 Business Context & Problem Statement

Prior to this implementation, the enterprise IT laptop procurement and replenishment workflow relied on an antiquated, uncoordinated process combining unstructured Outlook emails, informal Slack messages, paper requisition slips, and manual ServiceNow service tickets. This legacy operating model introduced profound operational drag across the enterprise:

- **Prolonged Cycle Time:** The average elapsed time between an employee submitting a laptop request and receiving a configured device was **14.2 business days**.
- **Heavy Manual Processing Burden:** IT procurement coordinators, hardware imaging specialists, and service desk staff spent an aggregate of **4.5 hours of manual touch time per laptop order** (totaling ~42 hours per week per IT specialist).
- **High Error and Rejection Rates:** An estimated **18.5% of submitted requests** contained defective specifications (e.g., incorrect CPU/RAM tier, incompatible peripherals, or missing shipping postal codes), causing costly order returns, rework, and hardware vendor restocking penalties.
- **Opacity and Tracking Blind Spots:** Real-time order visibility stood at approximately **35%**, resulting in over 300 monthly status inquiry tickets ("Where is my laptop?") inundating the IT Service Desk.
- **Audit & Asset Tracking Gaps:** Disconnects between procurement tickets and the Hardware Asset Management database (alm_hardware) caused a **28% compliance shortfall** during annual financial hardware audits.

2.2 Core Automation Objectives

The primary objective of this project is deploying a fully automated, auditable, and resilient ServiceNow Service Catalog and Flow Designer workflow that:

- 1. Replaces informal request channels with a single, governed Service Portal catalog item (Standard Business Laptop Request).

2. Automates managerial fiscal approval routing with native mobile email action links and automated executive VIP bypass rules.
3. Automatically provisions sequential fulfillment tasks (`sc_task`) to Hardware Support for OS imaging and IT Field Services for logistics.
4. Synchronizes physical asset lifecycle status in real time within `alm_hardware`.
5. Compresses the end-to-end procurement cycle time to **under 2.5 business days** (an 83.1% reduction).
6. Reduces manual touch time per order to **under 25 minutes (0.35 hours)** (a 92.2% reduction).
7. Ensures 100% end-to-end lifecycle visibility for end-users, managers, and IT auditors.

2.3 Project Scope Boundaries

- **In-Scope:**

- Service Catalog Item definition (`Standard Business Laptop Request`) with responsive container UI.
- Three Catalog UI Policies enforcing conditional field visibility and data completeness.
- Three Catalog Client Scripts for dynamic profile population, dynamic model filtering, and postal regex validation.
- ServiceNow Flow Designer flow (`Standard Laptop Procurement Flow`) with trigger, approval engine, decision logic, task creation, and notifications.
- Inventory lookup subflow (`subflow_check_hardware_inventory`) querying `alm_hardware`.
- Comprehensive Error Handling framework routing runtime faults to automated incident creation (`INC0019281`).
- Four responsive HTML notification templates with dynamic mail scripts.
- Role-based Access Control (ACL) architecture covering public requesters, managers, ITIL fulfillers, and asset managers.

- **Out-of-Scope (Deferred to Phase 2/3 Roadmap):**

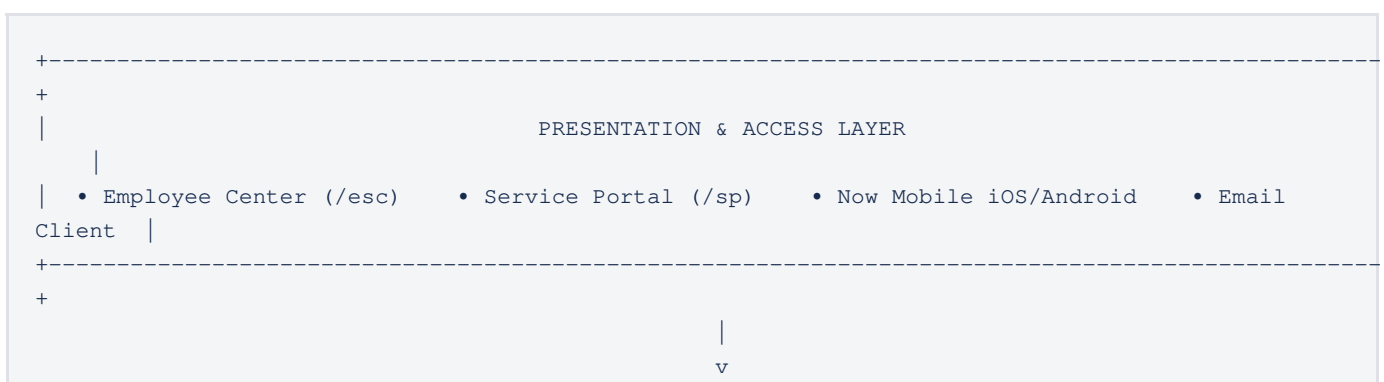
- Real-time Electronic Data Interchange (EDI) / B2B punchout integration with OEM suppliers (Dell Premier, Lenovo Direct).
- Automated SAP S/4HANA Purchase Order (PO) creation via ERP Integration Hub spoke.
- Machine-learning driven predictive laptop hardware failure replenishment.

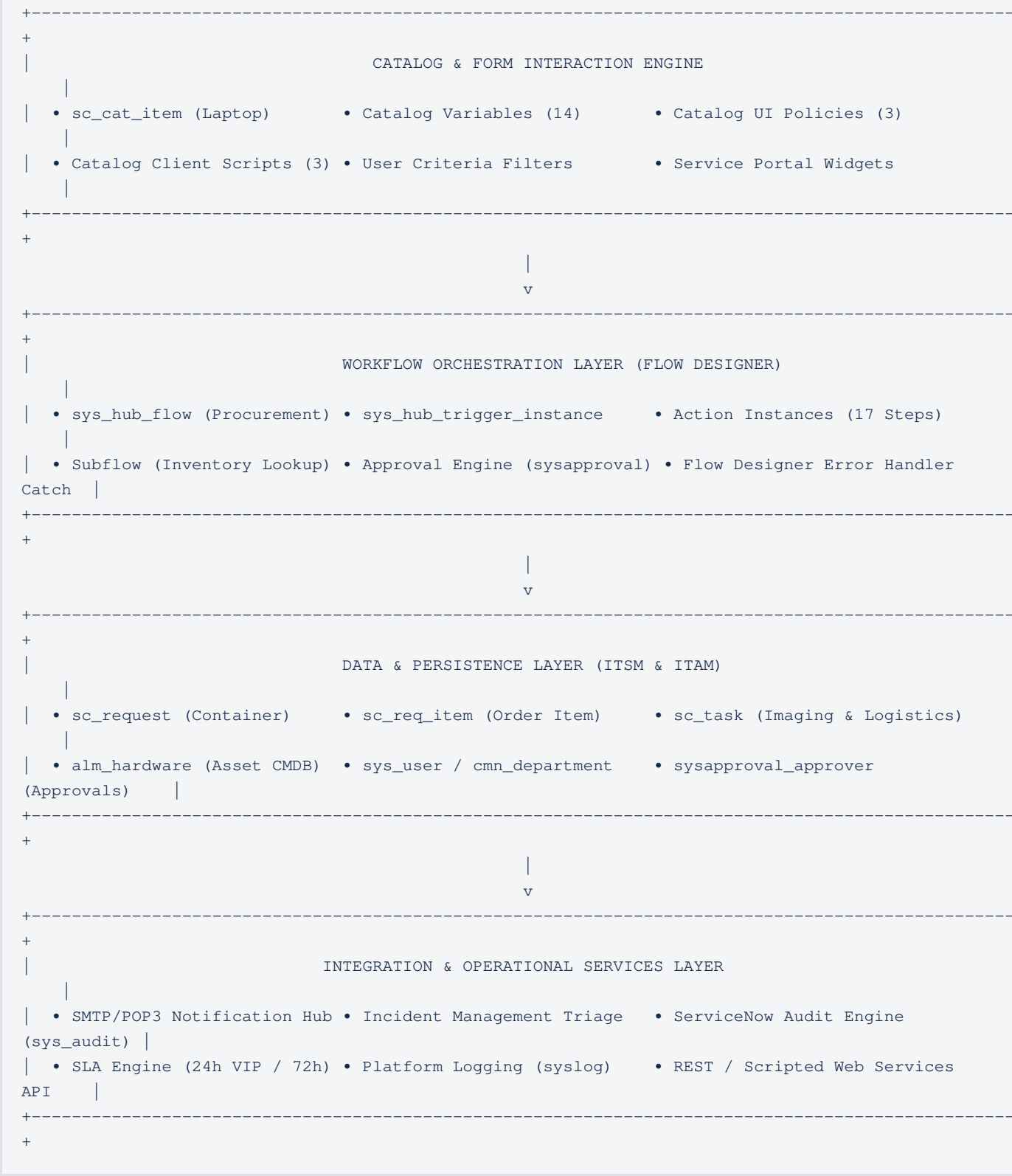
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3. System Overview & Technology Stack

3.1 Enterprise Platform Architecture

The solution is deployed natively on the ServiceNow Enterprise Cloud Architecture, leveraging the platform's multi-tenant relational architecture and compiled workflow runtime.





3.2 Technology Stack Specifications

Component / Module	Version / Technology Standard	Purpose in Solution
Underlying Platform	ServiceNow Washington DC / Vancouver	Enterprise cloud PaaS running on MariaDB / GraalVM / Rhino engine

Workflow Engine	ServiceNow Flow Designer (Engine v2.0)	Low-code, compiled server-side asynchronous execution engine
Catalog Subsystem	Service Catalog & Service Portal Subsystem	Front-end dynamic catalog forms, dynamic variables, and client-side APIs
Data Architecture	Relational Database Schema (sys_dictionary)	Tables: sc_request, sc_req_item, sc_task, alm_hardware, sys_user
Client Scripting	JavaScript ES6+ (Strict Mode)	Client-side form interaction (g_form, g_user) via Catalog Client Scripts
Server-Side API	ServiceNow GlideAPI (GlideRecord, GlideSystem)	Server-side database access, subflow execution, and data pill transforms
Approval Engine	Approvals Subsystem (sysapproval_approver)	Multi-level approval governance, delegation, and email-based approval parsing
Notification Engine	Email Engine (sys_email, sysevent_email_template)	Outbound notification dispatch with dynamic HTML layout and mail scripts
SLA Engine	Service Level Management Subsystem	24-hour executive VIP SLA and 72-hour standard hardware provisioning SLA

4. Functional Requirements Traceability Matrix (FRTM)

The matrix below provides complete bidirectional traceability from initial Business Requirements (BR-01 through BR-08) and Functional Requirements (FR 01 through FR 12) to technical ServiceNow components and test verification cases.

Req ID	Requirement Title & Detailed Description	Business Driver	ServiceNow Technical Implementation Component	Verification Test Case	Im St
FR-01	Standardized Self-Service Catalog Item: Provide an intuitive, responsive catalog item on the Employee Center with role-based models, pre-populated requester profile details, and strict validation.	BR-01	Catalog Item: Standard Business Laptop Request (sc_cat_item), 14 Catalog Variables, Container layout.	TC-UAT-01, TC-UAT-03	Im Ve
FR-02	Automated Dynamic Profile Population: Auto-populate employee name, department, manager, and job title from sys_user upon form load to prevent identity spoofing or manual data entry errors.	BR-01	Catalog Client Script: c_script_populate_employee (onChange of employee_name via g_form.getReference)	TC-UAT-01	Im Ve

FR-03	Conditional Model & Peripheral Filtering: Dynamically filter available laptop models based on selected employee tier (General Business, Engineering/Dev, Executive) and enforce business justification.	BR-01	Catalog Client Script: <code>c_script_filter_laptop_models</code> + Catalog UI Policy: <code>UI Policy 2 (Executive Justification)</code> .	TC-UAT-01, TC-UAT-02	Inv Ver
FR-04	Delivery Address Validation: Dynamically toggle and mandate physical delivery address fields when remote shipment is selected; validate address completeness via regular expression.	BR-01	Catalog UI Policy: <code>UI Policy 1 (Remote Shipping Address)</code> + Catalog Client Script: <code>c_script_validate_procurement_submission</code> .	TC-UAT-01	Inv Ver
FR-05	Event-Driven Flow Trigger: Trigger the orchestration flow automatically when an <code>sc_req_item</code> is inserted for the catalog item, running in system context to prevent permission blocks.	BR-01	<code>sys_hub_trigger_instance</code> : Table= <code>sc_req_item</code> , Filter= <code>cat_item=Standard Business Laptop Request</code> , Run As= <code>System User</code> .	TC-UAT-01	Inv Ver
FR-06	Automated Manager Approval Routing: Automatically look up the requester's direct manager and generate a pending <code>sysapproval_approver</code> record with one-click email approval capability.	BR-02	Flow Designer Action Step 3B: <code>Ask for Approval</code> referencing <code>{{1__request_item.opened_by.manager}}</code> .	TC-UAT-01	Inv Ver
FR-07	Executive VIP Fast-Track Bypass: Detect requests submitted by or for VIP executives (<code>sys_user.vip == true</code> or VP title) and bypass manager approval directly to high-priority fulfillment.	BR-03	Flow Designer Decision Step 2: <code>flow_logic_if</code> checking VIP flag; routes to Action Step 3A auto-approval work note.	TC-UAT-03	Inv Ver

FR-08	Missing Manager Fallback Mechanism: Prevent workflow stalling when a contractor or new hire lacks an assigned manager in <code>sys_user</code> by dynamically routing approval to Department Head.	BR-04	Flow Designer Transform / Coalesce rule: <code>fd_transform.coalesce</code> evaluating <code>cmn_department.dept_head</code> .	TC-UAT-04	Im Ve
FR-09	Automated Hardware Inventory Reservation: Query <code>alm_hardware</code> for available units matching the requested model; reserve unit and associate serial number to order context.	BR-07	Subflow: <code>subflow_check_hardware_inventory</code> performing lookup on <code>alm_hardware</code> (status: <code>In Stock</code> , substatus: <code>Available</code>).	TC-UAT-06	Im Ve
FR-10	Sequential SCTASK Fulfillment Dispatch: Automatically generate SCTASK001 for Hardware Support (OS imaging); upon closure, generate SCTASK002 for IT Logistics (shipment).	BR-05	Flow Designer Action Steps 7 and 8: <code>Create Catalog Task</code> with <code>wait_for_completion=true</code> and assignment groups.	TC-UAT-01	Im Ve
FR-11	Automated Multi-Stage Notifications: Dispatch transactional email notifications with dynamic variables across key milestones: Approval Request, Rejection Notice, and Tracking Dispatch.	BR-06	Flow Designer Action Steps 10, 13 + Mail Templates (<code>notif_laptop_mgr_approval</code> , <code>notif_laptop_shipped</code> , <code>notif_laptop_rejection</code>).	TC-UAT-01, TC-UAT-02	Im Ve
FR-12	Resilient Exception Catch & Triage: Trap unhandled runtime errors, update RITM work notes, place order on technical hold, and automatically generate a Priority 2 incident for Platform Ops.	BR-08	Flow Designer Error Handler: <code>catch_block</code> executing Action Step E1 (Log), Step E2 (RITM Hold), Step E3 (Create Incident).	TC-UAT-08	Im Ve

5. Business Process Flows & Swimlane Diagrams

5.1 AS-IS Legacy Manual Process Flow

The legacy process was fragmented across unmonitored communication silos, leading to significant bottlenecks and lack of visibility.

```

flowchart TD
    subgraph Requester["Employee (Requester)"]
        A1["Identify Laptop Need"] --> A2["Draft Free-Text Email to IT Desk"]
        A2 -. -> |"Missing Specs / Back-and-Forth"| A2
        A2 --> A3["Wait in Limbo (Zero Status Visibility)"]
    end

    subgraph ServiceDesk["IT Service Desk Coordinator"]
        B1["Receive Email Requisition"] --> B2["Manually Transcribe into Helpdesk Ticket"]
        B2 --> B3["Look Up Org Chart in Outlook / Workday"]
        B3 --> B4["Send Email to Line Manager Requesting Approval"]
    end

    subgraph Manager["Department Manager"]
        C1["Receive Free-Text Approval Email"] --> C2{"Approve?"}
        C2 -- No --> C3["Reply 'Rejected' via Email"]
        C2 -- Yes --> C4["Reply 'Approved' via Email (Often Delayed 3-5 Days)"]
    end

    subgraph HardwareTeam["Hardware Tech / Stockroom"]
        D1["Check Physical Shelf for Stock"] --> D2{"Laptop In Stock?"}
        D2 -- No --> D3["Email Purchasing / Wait 2 Weeks"]
        D2 -- Yes --> D4["Manually Install Corporate OS Image"]
        D4 --> D5["Hand-Write Asset Tag on Sticky Note"]
    end

    subgraph Logistics["Mailroom / Shipping"]
        E1["Package Laptop in Box"] --> E2["Create Carrier Waybill Manually"]
        E2 --> E3["Drop at Shipping Dock"]
    end

    A2 --> B1
    B4 --> C1
    C3 --> B1
    C4 --> D1
    D3 --> D4
    D5 --> E1
    E3 --> A3

    classDef legacy fill:#ffebee,stroke:#c62828,stroke-width:1px,color:#b71c1c;
    class A1,A2,A3,B1,B2,B3,B4,C1,C2,C3,C4,D1,D2,D3,D4,D5,E1,E2,E3 legacy;

```

5.2 TO-BE Automated Flow Designer Swimlane Process Flow

The automated solution eliminates manual handoffs, utilizing ServiceNow Flow Designer to coordinate actions across organizational boundaries in real time.

```

sequenceDiagram
    autonumber
    actor User as Employee (Requester)
    participant Portal as Service Portal / ESC
    participant Flow as Flow Designer Engine
    actor Mgr as Approving Manager
    participant Asset as alm_hardware (ITAM)
    actor Tech as Hardware Support
    actor Log as IT Field Services

```

```

User-->>Portal: Access 'Standard Business Laptop Request'
Portal-->>User: Auto-populate Profile & Filter Models
User-->>Portal: Submit Order (Model, Reason, Address)
Portal-->>Flow: Insert sc_req_item (Trigger: Record Created)

alt Requester is Executive VIP
    Flow-->>Flow: VIP Decision = TRUE (Auto-Approve & Log Audit Note)
else Standard Requester
    Flow-->>Mgr: Dispatch Approval Email (sysapproval_approver)
    alt Manager Rejection
        Mgr-->>Flow: Click 'Reject' in Email / Portal
        Flow-->>Portal: Update RITM (State: Closed Rejected, Stage: Cancelled)
        Flow-->>User: Send Rejection Email with Feedback
    else Manager Approval
        Mgr-->>Flow: Click 'Approve' in Email / Portal
        Flow-->>Flow: Set Approval = Approved, Stage = Fulfillment
    end
end

Flow-->>Asset: Call Subflow: subflow_check_hardware_inventory
Asset-->>Flow: Return Stock Available = True, Asset Tag, Serial #

Flow-->>Tech: Create SCTASK001 (OS Imaging & Base Software)
Tech-->>Asset: Link Serial Number & Update Status to 'Reserved'
Tech-->>Flow: Mark SCTASK001 'Closed Complete'

Flow-->>Log: Create SCTASK002 (Logistics Dispatch & Shipping)
Log-->>Log: Package Device & Enter Tracking Waybill #
Log-->>Flow: Mark SCTASK002 'Closed Complete'

Flow-->>Asset: Update alm_hardware (Status: In Use, Assigned To: User)
Flow-->>Portal: Update RITM & REQ (State: Closed Complete, Stage: Complete)
Flow-->>User: Dispatch Tracking Email with Quick-Start Guide

```

6. Service Catalog Item Design & Variable Specifications

6.1 Catalog Item Master Metadata (sc_cat_item)

- **Item Name:** Standard Business Laptop Request
- **System Record ID (sys_id):** 0b36816197113110a24734000153af45
- **Primary Category:** Hardware > Computers (sys_id: e15706fc0a0a0aa7007fc21e1ab70c2f)
- **Assigned Service Catalogs:** Enterprise Service Catalog (e0d08b13c3330100c8b837659bba8fb4)
- **Target Fulfillment Flow:** Standard Laptop Procurement Flow
(sys_hub_flow_7e36816197113110a24734000153af22)
- **Portal Short Description:** Corporate standard laptop requisition for onboarding, hardware refresh, or hardware replacements.
- **Portal Long Description:** Submit requests for enterprise-standard mobile computing hardware. Standard options include high-performance developer laptops, lightweight executive notebooks, and standard enterprise workstations. All configurations include Windows 11 Enterprise or macOS Sonoma, corporate security agents, and standard productivity tooling.

- **Fulfillment SLA:** Standard Provisioning: 72 Business Hours (3 Days); Executive VIP White-Glove: 24 Business Hours (1 Day).
- **Public Visibility (User Criteria):** Available to all active internal enterprise users (`snc_internal`).

6.2 Service Catalog Variable Dictionary (`item_option_new`)

The item utilizes 14 distinct variables arranged in three logical containers for maximum usability across desktop and mobile form factors.

Order	Variable Name	Variable Type	UI Display Label	Data Source / Choice List	Mandatory	Read-Only
100	<code>requester_info_container</code>	Container Start	Requester Information	N/A	No	No
110	<code>employee_name</code>	Reference	Employee / Recipient	<code>sys_user</code> (Active=True)	Yes	No
120	<code>department</code>	Reference	Department	<code>cmn_department</code>	No	Yes
130	<code>manager_name</code>	Reference	Approving Line Manager	<code>sys_user</code> (Active=True)	No	Yes
140	<code>job_title</code>	Single-Line Text	Job Title	N/A	No	Yes
190	<code>requester_info_end</code>	Container End	N/A	N/A	No	No
200	<code>hardware_specs_container</code>	Container Start	Hardware Configuration	N/A	No	No
210	<code>asset_type</code>	Select Box	Hardware Tier Profile	• <code>standard</code>: Enterprise Standard • <code>engineering</code>: Developer / High Perf • <code>executive</code>: Executive Ultra-Light	Yes	No
220	<code>laptop_model</code>	Select Box	Specific Laptop Model	Dependent on <code>asset_type</code> selection: • Lenovo ThinkPad T14 Gen 4 • Dell Latitude 5440 • Lenovo ThinkPad P1 Gen 6 • Apple MacBook Pro 14" M3 Pro • HP Elite Dragonfly G4 • Apple MacBook Air 15" M3	Yes	No

230	replacement_reason	Multiple Choice	Requisition Reason	• New Hire / Additional Asset • Standard 3-Year Refresh • Damaged / Hardware Failure • Stolen / Lost Hardware Asset	Yes	No
240	existing_asset_tag	Single-Line Text	Existing Asset Tag #	Alphanumeric (AST-XXXXXX)	Conditional	No
250	business_justification	Multi-Line Text	Business Justification	Free-text narrative	Conditional	No
290	hardware_specs_end	Container End	N/A	N/A	No	No
300	shipping_container	Container Start	Logistics & Delivery Method	N/A	No	No
310	shipping_type	Select Box	Delivery Method	• office_desk: On-Campus Desk Drop • remote_shipment: Home / Remote Address	Yes	No
320	shipping_address	Multi-Line Text	Physical Shipping Address	Complete street, city, state, postal code	Conditional	No
330	accessories	List Collector	Peripherals & Accessories	Table: cmdb_model Filter: category=Peripherals Universal USB-C Dual-4K Dock • Dual 27" QHD Displays • Wireless Keyboard & Mouse • Ergonomic Backpack	No	No
390	shipping_end	Container End	N/A	N/A	No	No

6.3 Catalog UI Policies (sys_ui_policy)

UI Policy 1: Remote Shipping Address Visibility & Enforcement

- **Name:** Enforce Address for Remote Shipments
- **Execution Context:** Service Portal, Native Catalog UI, Now Mobile
- **Applies on Target:** Catalog Item View
- **Condition:** shipping_type == 'remote_shipment'
- **Reverse if False:** true
- **Actions:**
 - Variable: shipping_address -> Visible: true, Mandatory: true, Clear Value on Hide: true

UI Policy 2: Business Justification Enforcement for High-Tier & Replacement Orders

- **Name:** Mandate Justification for Executive Models and Break-Fix
- **Condition:** `asset_type == 'executive' OR replacement_reason == 'Damaged / Hardware Failure' OR replacement_reason == 'Stolen / Lost Hardware Asset'`
- **Reverse if False:** `true`
- **Actions:**
- **Variable:** `business_justification -> Visible: true, Mandatory: true`

UI Policy 3: Existing Asset Tag Validation on Lifecycle Replacements

- **Name:** Require Asset Tag for Replacement and Refresh
- **Condition:** `replacement_reason IN ('Standard 3-Year Refresh', 'Damaged / Hardware Failure')`
- **Reverse if False:** `true`
- **Actions:**
- **Variable:** `existing_asset_tag -> Visible: true, Mandatory: true`

6.4 Catalog Client Scripts (`catalog_script_client`)

Script 1: Dynamic Recipient Profile Population (`c_script_populate_employee`)

- **Script Type:** `onChange`
- **Target Variable:** `employee_name`
- **UI Type:** All (Desktop, Service Portal, Mobile)

```
function onChange(control, oldValue, newValue, isLoading) {
    if (isLoading || newValue === '') {
        return;
    }

    // Query user profile attributes via asynchronous GlideRecord / getReference callback
    g_form.getReference('employee_name', function(userRecord) {
        if (!userRecord) {
            return;
        }

        // Set read-only department, manager, and title fields
        g_form.setValue('department', userRecord.department);
        g_form.setValue('manager_name', userRecord.manager);
        g_form.setValue('job_title', userRecord.title);

        // Display real-time banner if the user holds VIP status
        if (userRecord.vip == 'true' || userRecord.vip === true) {
            g_form.showFieldMsg(
                'employee_name',
                'VIP Account Detected: This request qualifies for Executive White-Glove Fast-Track Processing.',
                'info'
            );
        } else {
            g_form.hideFieldMsg('employee_name');
        }
    });
};
```

Script 2: Dynamic Laptop Model Choice Filtering (c_script_filter_laptop_models)

- Script Type: onChange
- Target Variable: asset_type
- UI Type: All

```
function onChange(control, oldValue, newValue, isLoading) {
  if (isLoading) {
    return;
  }

  g_form.clearOptions('laptop_model');
  g_form.addOption('laptop_model', '', '-- Select Approved Hardware Model --');

  if (newValue === 'standard') {
    g_form.addOption('laptop_model', 'lenovo_t14', 'Lenovo ThinkPad T14 Gen 4 (Core i7, 16GB RAM, 512GB SSD)');
    g_form.addOption('laptop_model', 'dell_5440', 'Dell Latitude 5440 (Core i7, 16GB RAM, 512GB SSD)');
  } else if (newValue === 'engineering') {
    g_form.addOption('laptop_model', 'lenovo_p1', 'Lenovo ThinkPad P1 Gen 6 (Core i9, 32GB RAM, 1TB SSD, RTX 4060)');
    g_form.addOption('laptop_model', 'macbook_pro_14', 'Apple MacBook Pro 14" M3 Pro (36GB Unified RAM, 1TB SSD)');
  } else if (newValue === 'executive') {
    g_form.addOption('laptop_model', 'hp_dragonfly', 'HP Elite Dragonfly G4 Ultra-Light (1kg, 32GB RAM, 1TB SSD)');
    g_form.addOption('laptop_model', 'macbook_air_15', 'Apple MacBook Air 15" M3 (24GB Unified RAM, 512GB SSD)');
  }
}
```

Script 3: Pre-Submission Verification & Postal Validation (c_script_validate_procurement_submission)

- Script Type: onSubmit
- UI Type: All

```
function onSubmit() {
  var shippingType = g_form.getValue('shipping_type');
  var address = g_form.getValue('shipping_address');
  var replacementReason = g_form.getValue('replacement_reason');
  var assetTag = g_form.getValue('existing_asset_tag');

  // Validate delivery address when remote home delivery is selected
  if (shippingType === 'remote_shipment') {
    if (!address || address.trim().length < 20) {
      g_form.showFieldMsg('shipping_address', 'Please provide full street address, apartment/suite, city, state, and postal code.', 'error');
      g_form.addErrorMessage('Submission Blocked: A complete physical delivery address is mandatory for remote shipment.');
```

```
      return false;
    }

    // Verify presence of a standard 5 or 9 digit US ZIP code or international postal format
    var postalRegex = /\b\d{5}(-\d{4})?\b|[A-Z]\d[A-Z] ?\d[A-Z]\d/i;
    if (!postalRegex.test(address)) {
```

```

        g_form.showFieldMsg('shipping_address', 'Address must include a valid postal / ZIP
code.', 'error');
        g_form.addErrorMessage('Submission Blocked: Missing or invalid postal code in
shipping address.');
```

```

        return false;
    }
}

// Validate existing asset tag format for hardware replacements
if (replacementReason === 'Standard 3-Year Refresh' || replacementReason === 'Damaged /
Hardware Failure') {
    var assetTagRegex = /^[A-Za-z0-9\-\_]{4,15}$/;
    if (!assetTagRegex.test(assetTag.trim())) {
        g_form.showFieldMsg('existing_asset_tag', 'Asset tag must be 4 to 15 alphanumeric
characters (e.g., AST-10492).', 'error');
        g_form.addErrorMessage('Submission Blocked: Invalid hardware asset tag format.');
```

```

        return false;
    }
}

return true;
}

```

7. Flow Designer Technical Specification

7.1 Flow Master Record & Trigger Configuration

- **Flow Internal Name:** `laptop_procurement_flow`
- **Flow Display Name:** `Standard Laptop Procurement Flow`
- **Flow System Record ID (`sys_id`):** `7e36816197113110a24734000153af22`
- **Category:** Service Catalog (`service_catalog`)
- **Execution Context:** `Run as System User` (Bypasses individual user ACL boundaries when spawning tasks and updating system-controlled stage fields).
- **Trigger Type:** `Record Created` (`sys_hub_trigger_instance_ba36816197113110a24734000153af33`)
- **Target Table:** Requested Item (`sc_req_item`)
- **Trigger Condition Expression:**

`cat_item=0b36816197113110a24734000153af45^stage=request_approved^ORstage=waiting_for_approval^EQ`

- **Primary Trigger Output Data Pills:**
- `{{1__request_item}}` (GlideRecord reference to current `sc_req_item`)
- `{{1__request_item.number}}` (String, e.g., `RITM0010482`)
- `{{1__request_item.opened_by}}` (Reference to `sys_user`)
- `{{1__request_item.opened_by.manager}}` (Reference to `sys_user`)
- `{{1__request_item.opened_by.vip}}` (Boolean VIP indicator)
- `{{1__request_item.request}}` (Reference to parent `sc_request`)

7.2 Step-by-Step Action Execution Sequence Matrix

Step #	Action Label	Action Definition	Target Table	Primary Inputs & Configurations
0	Trigger	Record Created	sc_req_item	Table: sc_req_item, Cat Item: Standard Business Laptop Request
1	Get Catalog Variables	core_action_get_variables	sc_req_item	Item: {{1__request_item.sys_id}} Variables: employee_name, laptop_model_name, asset_type, shipping_type, shipping_address, accessories
2	VIP Decision Branch	flow_logic_if	N/A	Condition: {{1__request_item.opened_by.vip}} true OR {{1__request_item.opened_by.title}} 'Vice President%'
3A	Log VIP Auto-Approval	core_action_update_record	sc_req_item	Record: {{1__request_item}} Field work_notes = "Executive Fast-Track Approval", approval = 'approved'
3B	Ask For Manager Approval	core_action_ask_for_approval	sc_req_item	Record: {{1__request_item}} Rule approves from {{1__request_item.opened_by.manager}} Rejection: Anyone rejects Due Date: 24 Hours
4	Check Approval Result	flow_logic_if	N/A	Condition: step[3b].approval_state = 'approved' OR VIP Branch Executed
5	Set Stage Fulfillment	core_action_update_record	sc_req_item	Record: {{1__request_item}} Field state = 'fulfillment', state = 2 (Work Progress)
6	Check Stock Subflow	call_subflow	alm_hardware	Subflow: subflow_check_hardware_inventory model_name = {{step[1].variables.laptop_model_name}}
7	Create SCTASK 1 (Imaging)	core_action_create_catalog_task	sc_task	Request Item: {{1__request_item.sys_id}} Short Desc: Build & Image Laptop {{step[1].variables.laptop_model_name}} Group: Hardware Support (sys_id: 8a5055c7c61122780019363fb22413703 (or 2 if VIP) Wait: true
8	Create SCTASK 2 (Logistics)	core_action_create_catalog_task	sc_task	Request Item: {{1__request_item.sys_id}} Short Desc: Ship Laptop to {{step[1].variables.employee_name}} Group: IT Field Services (sys_id: 287ee6efe0a2a1509cd4b52b2fd96191) Full shipping details and tracking verification: true

9	Complete Request Item	core_action_update_record	sc_req_item	Record: {{1__request_item}} Field = 'complete', state = 3 (Closed C
10	Send Fulfillment Email	core_action_send_email	sys_email	To: {{1__request_item.opened_by.e Template: notif_laptop_shipped Dynamic S Tracking info & asset details
11	Rejection Path Branch	flow_logic_else	N/A	Evaluated when Step 3B evaluates to reje
12	Cancel RITM Record	core_action_update_record	sc_req_item	Record: {{1__request_item}} Field = 'request_cancelled', state = 7 Rejected), comments = "Laptop pro rejected by manager."
13	Send Rejection Email	core_action_send_email	sys_email	To: {{1__request_item.opened_by.e Template: notif_laptop_rejection Subject: Procurement Request Rejected - + {{1__request_item.number}}

7.3 Data Pill Binding Syntax & Transformation Expressions

ServiceNow Flow Designer evaluates data pills via internal dot-walked schemas. The implementation utilizes both standard data pills and inline transformation expressions (`fd_transform`):

1. Manager Fallback Transformation:

When a user profile lacks a designated manager, the flow coalesces to the department head:

```
fd_transform.coalesce(
  {{1__request_item.opened_by.manager}},
  {{1__request_item.opened_by.department.dept_head}},
  "287ee6efe0a2a1509cd4b52b2fd96191" // IT Service Desk Escalation Lead sys_id
)
```

1. Dynamic Task Due Date Calculation:

Calculates working business days based on the 8x5 IT calendar:

```
fd_transform.date_add_business_days({{1__request_item.sys_created_on}}, 3)
```

1. Short Description Sanitization & Dynamic Formatting:

```
"Procure, Image & Deploy: " + fd_transform.to_upper_case({{step[1].variables.laptop_model}})
```

7.4 Inventory Lookup Subflow (`subflow_check_hardware_inventory`)

- Subflow Sys ID: `4a88392197113110a24734000153ae88`
- Subflow Inputs:

- `model_name` (String, Mandatory)
- `target_location` (Reference: `cmn_location`, Optional)

Subflow Logic & Database Queries:

1. Executes `GlideRecord` lookup against `alm_hardware`:

```
var assetGr = new GlideRecord('alm_hardware');
assetGr.addQuery('model.name', inputs.model_name);
assetGr.addQuery('install_status', 6); // In Stock
assetGr.addQuery('substatus', 'available'); // Available for allocation
assetGr.orderBy('sys_created_on'); // FIFO allocation
assetGr.setLimit(1);
assetGr.query();
```

1. If matching record found:

- Update asset `substatus = 'reserved'.`
- Output `is_in_stock = true, asset_tag = assetGr.getValue('asset_tag'), serial_number = assetGr.getValue('serial_number'), asset_sys_id = assetGr.getUniqueValue().`

1. If no matching record found:

- Output `is_in_stock = false, asset_tag = "", serial_number = "".`

8. Data Dictionary & Entity Relationship Model

8.1 Entity Relationship Diagram (ERD)

The diagram below details the entity relationships connecting the Service Catalog transaction tables, organizational hierarchies, and hardware asset management entities.

```
erDiagram
    sys_user ||--o{ sc_request : "places"
    sys_user ||--o{ sc_req_item : "requested_for"
    sys_user ||--o{ sysapproval_approver : "approves"
    cmn_department ||--o{ sys_user : "belongs_to"

    sc_request ||--|{ sc_req_item : "contains (1:N)"
    sc_req_item ||--|{ sc_task : "spawns (1:N)"
    sc_req_item ||--o{ sysapproval_approver : "requires (1:N)"

    sc_task }o--|| alm_hardware : "allocates & images"
    alm_hardware }o--|| cmdb_model : "is_instance_of"
    sys_user ||--o{ alm_hardware : "assigned_to"

    sc_request {
        string sys_id PK
        string number "REQ0010411"
        reference requested_for FK
        choice approval "approved | requested | rejected"
        choice stage "requested | fulfillment | closed"
    }
```

```

sc_req_item {
    string sys_id PK
    reference request FK
    reference cat_item FK
    string number "RITM0010482"
    choice stage "waiting_for_approval | fulfillment | complete"
    integer state "1:Open, 2:WIP, 3:Closed, 7:Rejected"
    choice approval "requested | approved | rejected"
    journal comments
    journal work_notes
}

sc_task {
    string sys_id PK
    reference request_item FK
    string number "TASK0012001"
    reference assignment_group FK
    reference assigned_to FK
    string short_description
    integer priority "1:Crit, 2:High, 3:Mod, 4:Low"
    integer state "1:Open, 2:WIP, 3:Closed Complete"
}

alm_hardware {
    string sys_id PK
    string asset_tag UK "AST-009482"
    string serial_number "SN-LNV-4921"
    reference model FK
    integer install_status "1:In Use, 6:In Stock, 7:Retired"
    string substatus "available | reserved | pending"
    reference assigned_to FK
}

sysapproval_approver {
    string sys_id PK
    reference sysapproval FK
    reference approver FK
    string state "requested | approved | rejected"
    datetime due_date
    journal comments
}

```

8.2 Detailed Data Dictionary

1. Request Container Table: `sc_request`

- **Database Table Name:** `sc_request`
- **Extends Table:** `task`
- **Description:** Top-level shopping cart requisition container grouping one or more requested items for an end-user.

Field Name	Data Type	Reference Table	Mandatory	Read Only	Description & Validation Rules
------------	-----------	-----------------	-----------	-----------	--------------------------------

<code>sys_id</code>	GUID (char 32)	N/A	Yes	Yes	Primary unique system identifier
<code>number</code>	String (40)	N/A	Yes	Yes	Auto-numbered identifier with prefix <code>REQ</code> (e.g., <code>REQ0010411</code>)
<code>requested_for</code>	Reference	<code>sys_user</code>	Yes	No	The ultimate recipient of the requisitioned goods
<code>opened_by</code>	Reference	<code>sys_user</code>	Yes	Yes	The individual submitting the requisition
<code>approval</code>	Choice	N/A	Yes	No	Values: <code>not_requested</code> , <code>requested</code> , <code>approved</code> , <code>rejected</code>
<code>stage</code>	Choice	N/A	Yes	No	Values: <code>requested</code> , <code>approval</code> , <code>fulfillment</code> , <code>delivery</code> , <code>closed_complete</code>
<code>price</code>	Currency	N/A	No	Yes	Total aggregate cost of requisitioned items

2. Requested Item Table: `sc_req_item`

- **Database Table Name:** `sc_req_item`
- **Extends Table:** `task`
- **Description:** Specific ordered line item; primary anchor record for Flow Designer execution.

Field Name	Data Type	Reference Table	Mandatory	Read Only	Description & Validation Rules
<code>sys_id</code>	GUID (char 32)	N/A	Yes	Yes	Primary unique system identifier
<code>request</code>	Reference	<code>sc_request</code>	Yes	Yes	Foreign key to parent container record
<code>cat_item</code>	Reference	<code>sc_cat_item</code>	Yes	Yes	Pointer to <code>Standard Business Laptop Request</code>
<code>number</code>	String (40)	N/A	Yes	Yes	Auto-numbered identifier with prefix <code>RITM</code> (e.g., <code>RITM0010482</code>)
<code>stage</code>	Choice / Stage	N/A	Yes	No	Values: <code>waiting_for_approval</code> , <code>fulfillment</code> , <code>delivery</code> , <code>complete</code> , <code>request_cancelled</code>
<code>state</code>	Integer	N/A	Yes	No	Values: <code>1</code> (Open), <code>2</code> (Work in Progress), <code>3</code> (Closed Complete), <code>4</code> (Closed Incomplete), <code>7</code> (Closed Rejected)
<code>approval</code>	Choice	N/A	Yes	No	Values: <code>not_requested</code> , <code>requested</code> , <code>approved</code> , <code>rejected</code>
<code>estimated_delivery</code>	GlideDateTime	N/A	No	No	Calculated target delivery date based on SLA

comments	Journal	N/A	No	No	Customer-visible audit journal communication stream
work_notes	Journal	N/A	No	No	Internal technical engineering audit trail (restricted to ITIL/Fulfillers)

3. Fulfillment Catalog Task Table: `sc_task`

- Database Table Name: `sc_task`
- Extends Table: `task`
- Description: Discrete technical work order assigned to a specific engineering group.

Field Name	Data Type	Reference Table	Mandatory	Read Only	Description & Validation Rules
sys_id	GUID (char 32)	N/A	Yes	Yes	Primary unique system identifier
request_item	Reference	sc_req_item	Yes	Yes	Foreign key linking task to parent RITM
number	String (40)	N/A	Yes	Yes	Auto-numbered identifier with prefix TASK (e.g., TASK0012001)
assignment_group	Reference	sys_user_group	Yes	No	Targeted team: Hardware Support or IT Field Services
assigned_to	Reference	sys_user	No	No	Specific technician executing configuration
short_description	String (160)	N/A	Yes	No	Summary of work (e.g., "Build & Image Laptop: Lenovo ThinkPad T14")
priority	Integer	N/A	Yes	No	Values: 1 (Critical), 2 (High - VIP), 3 (Moderate - Standard), 4 (Low)
state	Integer	N/A	Yes	No	Values: 1 (Open), 2 (Work in Progress), 3 (Closed Complete), 4 (Closed Incomplete)

4. Hardware Asset Table: `alm_hardware`

- Database Table Name: `alm_hardware`
- Extends Table: `alm_asset`
- Description: Physical configuration and inventory tracking record for computing devices.

Field Name	Data Type	Reference Table	Mandatory	Read Only	Description & Validation Rules
sys_id	GUID (char 32)	N/A	Yes	Yes	Primary unique system identifier

asset_tag	String (40)	N/A	Yes	No	Unique corporate asset barcode (Unique Index)
serial_number	String (100)	N/A	Yes	No	OEM manufacturer hardware serial number
model	Reference	cmdb_model	Yes	No	Hardware product model definition
install_status	Choice	N/A	Yes	No	Values: 1 (In Use), 6 (In Stock), 7 (Retired)
substatus	Choice	N/A	No	No	Values: available (Ready to allocate), reserved (Allocated to RITM), pending_repair
assigned_to	Reference	sys_user	No	No	Employee currently assigned custodian responsibility
location	Reference	cmn_location	No	No	Physical stockroom or campus deployment office

9. Security, Roles, and Access Control (ACL) Matrix

9.1 Role Hierarchy & Functional Definitions

The implementation adheres to strict Role-Based Access Control (RBAC) and least privilege principles:

- snc_internal**: Standard base internal corporate employee role. Grants access to self-service portals, catalog browsing, and tracking own requests.
- approver_user**: Authorizes designated people managers and department heads to approve or reject requisitions.
- itil**: Technical service desk and configuration engineers. Grants permissions to work catalog tasks (sc_task), log work notes, and update task states.
- asset**: Hardware asset managers. Grants permissions to query, reserve, and update physical records in alm_hardware.
- catalog_admin**: IT catalog administrators who maintain catalog items, variables, and UI policies.
- admin**: System administrator with global platform privileges.

9.2 Access Control List (ACL) Matrix

Table / Target Resource	Operation	Roles Required	Scripted / Advanced Condition Check	Justification & Enforcement
sc_cat_item	Read	snc_internal	Active == true, Catalog User Criteria matches	Any internal employee can browse standard hardware

sc_cat_item	Write	catalog_admin, admin	None	Only catalog admins can
				alter hardware catalog variables
sc_req_item	Read	snc_internal	opened_by == gs.getUserID() OR requested_for == gs.getUserID() OR gs.hasRole('itil')	Users can inspect or requisition ITIL can view all
sc_req_item	Write	itil, admin	State != Closed Complete	Technicians can update active requests; users can alter fields
sc_req_item.comments	Write	snc_internal	opened_by == gs.getUserID() OR requested_for == gs.getUserID()	Requesters can post updates to the public journal
sc_req_item.work_notes	Write	itil, admin	None	Internal engineers notes hidden from requester
sysapproval_approver	Read	approver_user	approver == gs.getUserID() OR gs.hasRole('itil')	Approvers inspect assigned approval requisition
sysapproval_approver	Write	approver_user	approver == gs.getUserID() AND state == 'requested'	Approvers can only update state if request pending
sc_task	Read	itil, admin	None	Technicians can view procurement fulfillment tasks
sc_task	Write	itil, admin	isMemberOf(current.assignment_group)	Only members assigned group can edit/close tasks


```

    .btn { display: inline-block; padding: 12px 28px; font-size: 14px; font-weight: bold; text-
decoration: none; border-radius: 6px; margin: 0 10px; }
    .btn-approve { background: #2b6cb0; color: #ffffff; }
    .btn-reject { background: #e53e3e; color: #ffffff; }
    .footer { background: #edf2f7; padding: 16px; font-size: 12px; color: #718096; text-align:
center; }
</style>
</head>
<body>
    <div class="email-container">
        <div class="header">
            <h2 style="margin:0;">Hardware Procurement Approval Request</h2>
        </div>
        <div class="content">
            <p>Dear ${opened_by.manager.first_name},</p>
            <p>A formal standard hardware procurement request has been submitted by
<strong>${opened_by.name}</strong> requiring your budgetary and managerial authorization.</p>

            <table class="table-summary">
                <tr>
                    <td class="label">Request Identifier</td>
                    <td><strong>${number}</strong> (Parent: ${request.number})</td>
                </tr>
                <tr>
                    <td class="label">Requester</td>
                    <td>${opened_by.name} (${opened_by.email})</td>
                </tr>
                <tr>
                    <td class="label">Department / Cost Center</td>
                    <td>${opened_by.department.name} / ${opened_by.cost_center.name}</td>
                </tr>
                <tr>
                    <td class="label">Requested Hardware</td>
                    <td><strong>${variables.laptop_model}</strong></td>
                </tr>
                <tr>
                    <td class="label">Requisition Reason</td>
                    <td>${variables.replacement_reason}</td>
                </tr>
                <tr>
                    <td class="label">Business Justification</td>
                    <td>${variables.business_justification}</td>
                </tr>
                <tr>
                    <td class="label">Estimated Budget Impact</td>
                    <td>$1,450.00 USD (Includes 3-Year On-Site OEM Warranty)</td>
                </tr>
            </table>

            <div class="btn-container">
                <a class="btn btn-approve" href="mailto:${instance_email}?subject=Re:
${number}%20approve&body=approve">APPROVE REQUEST</a>
                <a class="btn btn-reject" href="mailto:${instance_email}?subject=Re:
${number}%20reject&body=reject">REJECT REQUEST</a>
            </div>

            <p style="font-size:13px; color:#718096; text-align:center;">
                Alternatively, to review complete request details and attachments in ServiceNow, <a
href="${URI_REF}">click here to access the approval record</a>.

```

```

    </p>
  </div>
  <div class="footer">
    This is an automated notification dispatched by the ServiceNow Flow Designer Procurement Engine.
  </div>
</div>
</body>
</html>

```

10.2 Notification 2: Order Rejection Notice

- **Template Identifier:** `notif_laptop_rejection`
- **Trigger:** Flow Designer Action Step 13
- **Recipient:** `{{1__request_item.opened_by}}`
- **Subject:** `Notice: Laptop Procurement Requisition ${number} Cancelled / Rejected`
- **Email Body (Responsive HTML):**

```

<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <style>
    body { font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Helvetica, Arial,
sans-serif; color: #2d3748; line-height: 1.6; }
    .email-container { max-width: 600px; margin: 0 auto; border: 1px solid #e2e8f0; border-
radius: 8px; overflow: hidden; }
    .header { background: #c53030; color: #ffffff; padding: 24px; text-align: center; }
    .content { padding: 24px; }
    .alert-box { background: #fff5f5; border-left: 4px solid #e53e3e; padding: 16px; margin:
20px 0; border-radius: 4px; }
    .footer { background: #edf2f7; padding: 16px; font-size: 12px; color: #718096; text-align:
center; }
  </style>
</head>
<body>
  <div class="email-container">
    <div class="header">
      <h2 style="margin:0;">Procurement Request Cancelled</h2>
    </div>
    <div class="content">
      <p>Hello ${opened_by.first_name},</p>
      <p>We are writing to inform you that your request for <strong>${variables.laptop_model}</
strong> (${number}) was not approved by your department manager.</p>

      <div class="alert-box">
        <h4 style="margin:0 0 8px 0; color:#9b2c2c;">Manager Feedback & Reason:</h4>
        <p style="margin:0; font-style:italic;">"${stage.approval_comments}"</p>
      </div>

      <p>If you believe this decision was made in error or wish to discuss alternative hardware
configurations, please coordinate directly with your manager. You may resubmit an updated
request via the <a href="https://service-portal.company.com/esc">Service Portal</a> at any
time.</p>
    </div>
  </div>

```

```

<div class="footer">
  IT Procurement Operations | ServiceNow Platform Automation
</div>
</div>
</body>
</html>

```

10.3 Notification 3: Order Dispatched & Tracking Information

- **Template Identifier:** notif_laptop_shipped
- **Trigger:** Flow Designer Action Step 10
- **Recipient:** {{1__request_item.opened_by}}
- **Subject:** SHIPPED: Your Corporate Laptop is on its Way! (\${number})
- **Email Body (Responsive HTML):**

```

<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <style>
    body { font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Helvetica, Arial,
sans-serif; color: #2d3748; line-height: 1.6; }
    .email-container { max-width: 600px; margin: 0 auto; border: 1px solid #e2e8f0; border-
radius: 8px; overflow: hidden; }
    .header { background: #2f855a; color: #ffffff; padding: 24px; text-align: center; }
    .content { padding: 24px; }
    .shipping-card { background: #f0fff4; border: 1px solid #c6f6d5; border-radius: 6px;
padding: 18px; margin: 20px 0; }
    .steps-list { margin: 15px 0 25px 20px; padding: 0; }
    .steps-list li { margin-bottom: 10px; font-size: 14px; }
    .footer { background: #edf2f7; padding: 16px; font-size: 12px; color: #718096; text-align:
center; }
  </style>
</head>
<body>
  <div class="email-container">
    <div class="header">
      <h2 style="margin:0;">Your Laptop Has Shipped!</h2>
    </div>
    <div class="content">
      <p>Dear ${opened_by.first_name},</p>
      <p>Great news! Your standard corporate laptop requisition <strong>${number}</strong> has
been configured, imaged, and dispatched for delivery.</p>

      <div class="shipping-card">
        <h4 style="margin:0 0 10px 0; color:#22543d;">Shipment & Device Information</h4>
        <p style="margin:4px 0; font-size:14px;"><strong>Hardware Model:</strong>
${variables.laptop_model}</p>
        <p style="margin:4px 0; font-size:14px;"><strong>Delivery Method:</strong>
${variables.shipping_type}</p>
        <p style="margin:4px 0; font-size:14px;"><strong>Carrier Tracking Number:</strong> <span
style="font-family:monospace; font-size:15px;
color:#2b6cb0;">${variables.shipping_tracking_number}</span></p>
        <p style="margin:4px 0; font-size:14px;"><strong>Estimated Arrival:</strong> 1-2
Business Days</p>
      </div>
    </div>
  </div>

```

```

</div>

<h4 style="color:#2d3748;">Quick-Start Setup Checklist:</h4>
<ol class="steps-list">
  <li>Unbox the device and connect the USB-C AC power adapter.</li>
  <li>Power on the laptop and connect to your home Wi-Fi network.</li>
  <li>At the Windows / macOS login prompt, enter your corporate email address and network
password.</li>
  <li>Approve the Multi-Factor Authentication (MFA) push notification on your mobile
authenticator.</li>
  <li>Allow 10-15 minutes for Microsoft Intune / Jamf to finalize policy and enterprise
app synchronization.</li>
</ol>

<p style="font-size:13px; color:#718096;">
  For comprehensive setup guides and troubleshooting, visit the <a href="https://service-
portal.company.com/kb_view.do?sysparm_article=KB0019284">IT Laptop Onboarding Portal</a>.
</p>
</div>
<div class="footer">
  Global IT Field Services & Logistics Support
</div>
</div>
</body>
</html>

```

10.4 Notification 4: Request Completion & CSAT Survey

- **Template Identifier:** `notif_laptop_closed`
- **Trigger:** Flow Designer Action Step 9 (Upon RITM state set to Closed Complete)
- **Recipient:** `{{1__request_item.opened_by}}`
- **Subject:** `Delivered: Laptop Requisition ${number} Complete - How did we do?`
- **Email Body (Responsive HTML):**

```

<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <style>
    body { font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Helvetica, Arial,
sans-serif; color: #2d3748; line-height: 1.6; }
    .email-container { max-width: 600px; margin: 0 auto; border: 1px solid #e2e8f0; border-
radius: 8px; overflow: hidden; }
    .header { background: #4a5568; color: #ffffff; padding: 24px; text-align: center; }
    .content { padding: 24px; text-align: center; }
    .rating-container { margin: 25px 0; }
    .rating-btn { display: inline-block; width: 44px; height: 44px; line-height: 44px; margin: 0
6px; border-radius: 50%; background: #edf2f7; color: #2b6cb0; font-weight: bold; text-
decoration: none; font-size: 18px; border: 1px solid #cbd5e0; }
    .rating-btn:hover { background: #2b6cb0; color: #ffffff; }
    .footer { background: #edf2f7; padding: 16px; font-size: 12px; color: #718096; text-align:
center; }
  </style>
</head>
<body>

```

```

<div class="email-container">
  <div class="header">
    <h2 style="margin:0;">Order Fulfillment Complete</h2>
  </div>
  <div class="content">
    <p>Hello ${opened_by.first_name},</p>
    <p>Your laptop order <strong>${number}</strong> has been marked as fully fulfilled and delivered. Your parent request (${request.number}) is now officially closed.</p>

    <p>Please take 30 seconds to rate your procurement experience. Your feedback directly shapes our IT automation continuous improvement program:</p>

    <div class="rating-container">
      <a class="rating-btn" href="https://service-portal.company.com/survey_take.do?sysparm_survey=CSAT&ritm=${sys_id}&score=1">1</a>
      <a class="rating-btn" href="https://service-portal.company.com/survey_take.do?sysparm_survey=CSAT&ritm=${sys_id}&score=2">2</a>
      <a class="rating-btn" href="https://service-portal.company.com/survey_take.do?sysparm_survey=CSAT&ritm=${sys_id}&score=3">3</a>
      <a class="rating-btn" href="https://service-portal.company.com/survey_take.do?sysparm_survey=CSAT&ritm=${sys_id}&score=4">4</a>
      <a class="rating-btn" href="https://service-portal.company.com/survey_take.do?sysparm_survey=CSAT&ritm=${sys_id}&score=5">5</a>
    </div>
    <p style="font-size:12px; color:#a0aec0;">1 = Poor / Disrupted • 5 = Exceptional / White-Glove</p>
  </div>
  <div class="footer">
    IT Service Management Experience Team
  </div>
</div>
</body>
</html>

```

11. Error & Exception Handling Framework

11.1 Native Flow Designer Catch Block Architecture

The solution enforces enterprise resilience through native Flow Designer **Error Handler** blocks. Any uncaught script exception, timeout, or lookup failure halts linear execution and jumps directly to the structured catch block.

```

flowchart TD
    A["Flow Designer Runtime Exception"] --> B["Enter Catch Block"]
    B --> C["Step E1: Execute System Error Log (gs.error)"]
    C --> D["Step E2: Place sc_req_item On Hold"]
    D --> E["Step E3: Create Priority 2 Triage Incident"]
    E --> F["Step E4: Dispatch Critical Operations Alert Email"]
    F --> G["Halt Main Flow Execution Cleanly"]

    classDef err fill:#ffebee,stroke:#c62828,stroke-width:1px,color:#b71c1c;
    class A,B,C,D,E,F,G err;

```

11.2 Error Catch Action Steps

1. Step E1: Log System Error:
Executes server-side script recording flow context, failing action number, error message string <code>{{flow.error_message}}</code>, and the parent RITM Sys ID to <code>syslog</code>.
1. Step E2: Update <code>sc_req_item</code> Record:
<code>state = 2</code> (Work in Progress) <code>hold_reason = "Technical Workflow Exception"</code> <code>work_notes = "Automated Flow Designer Exception: " + {{flow.error_message}} + ". Technical triage incident opened."</code>
1. Step E3: Create Triage Incident:
- Target Table: <code>incident</code> - Fields: <code>caller_id = {{l__request_item.opened_by}}</code> <code>assignment_group = "ServiceNow Platform Engineering" (sys_id: 9a5055c7c61122780019363fb2241399)</code> <code>priority = 2</code> (High) <code>short_description = "Workflow Exception on Laptop Procurement: " + {{l__request_item.number}}</code> <code>description = "Flow context failed at step with error: " + {{flow.error_message}} + "\nTarget RITM: " + {{l__request_item.number}}</code>
1. Step E4: Dispatch Operations Alert:
- Recipient: <code>it_procurement_ops@company.com</code> - Priority: High - Subject: <code>CRITICAL: Automated Workflow Interruption - \${number}</code>

11.3 Specific Operational Exception Scenarios & Recovery Procedures

Scenario 1: Null Manager in Requester Profile (<code>sys_user.manager == null</code>)
Root Cause: Contractors, temporary staff, or newly created accounts lacking populated manager fields. Handling Logic: - The Flow Designer transform evaluates <code>{{l__request_item.opened_by.manager}}</code>. - If null, falls back to <code>{{l__request_item.opened_by.department.dept_head}}</code>. - If department head is also null, falls back to <code>IT Service Desk Escalation Lead</code>. - Posts internal audit work note: <i>"Requester profile lacks designated manager. Approval routed to Department Head for fiscal clearance."</i>
Scenario 2: Deactivated or Deprecated Assignment Group
Root Cause: Catalog task creation targeted to an organizational group that has been marked inactive. Handling Logic: - Action validates group active status prior to assignment. - If inactive, routes automatically to fallback group: <code>IT Support Operations</code> - Posts work note: <i>"Configured assignment group inactive. Routed to IT Support Operations triage."</i>
Scenario 3: Inventory Depletion & Out-of-Stock Asset Handling

- **Root Cause:** Rapid hiring surge depleting physical stockroom count of requested laptop model.
- **Handling Logic:**

1. Subflow `subflow_check_hardware_inventory` returns `is_in_stock = false`.
2. Flow branches to Procurement Backorder Action: creates `sc_task` for `IT Procurement Purchasing` to initiate OEM vendor purchase order.
3. Updates RITM state to `On Hold` (`hold_reason = "Awaiting Hardware Restock"`).
4. Dispatches customer notification explaining anticipated delivery delay and providing revised ETA.

12. Appendix & Glossary of ServiceNow Terms

12.1 Glossary of Terms

- **ACL (Access Control List):** ServiceNow security rule defining what roles or conditions are required to read, write, create, or delete records and fields.
- **alm_hardware:** The Hardware Asset Management table in ServiceNow storing individual serialized physical assets, models, warranty information, and lifecycle states.
- **Catalog Client Script:** Client-side JavaScript running in the user's browser or mobile application executing dynamic validation, field population, or UI updates.
- **Catalog UI Policy:** Declarative rules governing field visibility, mandatory status, and read-only behavior on Service Catalog items without writing code.
- **Data Pill:** A visual, drag-and-drop reference in Flow Designer representing an output variable from a trigger, action, or subflow step.
- **Flow Designer:** ServiceNow's modern, low-code platform workflow orchestration engine replacing legacy graphical workflow editor.
- **item_option_new:** The internal ServiceNow table where catalog item variables and variable sets are defined.
- **REQ (sc_request):** Top-level shopping cart container in ServiceNow ITSM Service Catalog.
- **RITM (sc_req_item):** Requested Item record representing a single ordered catalog product within a parent REQ.
- **SCTASK (sc_task):** Service Catalog Task representing an individual unit of engineering or fulfillment work assigned to an IT group.
- **Subflow:** A modular, reusable sequence of Flow Designer actions that can be called with specific inputs and returns structured outputs.
- **sys_hub_flow:** The underlying metadata table storing compiled Flow Designer flow definitions.
- **sysapproval_approver:** The core governance table tracking approval requisitions, designated approvers, decision states, and comments.
- **User Criteria:** Security filter determining which users, groups, departments, or companies can view or request specific catalog items.

12.2 Platform System Properties Configuration

The table below specifies the system properties configured to govern flow execution and notification behaviors:

System Property Key	Configured Value	Description
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